

# Network Science

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## Lecture 07: Small-World Networks

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# Milgram's Experiment — The Question That Started It All

In 1967, the social psychologist Stanley Milgram mailed 296 letters to random people in Nebraska and Kansas. Each letter named a **target person** — a stockbroker in Boston — and asked the recipient to forward it to someone they knew *on a first-name basis* who might be "closer" to the target.

# Milgram's Experiment — The Question That Started It All

In 1967, the social psychologist Stanley Milgram mailed 296 letters to random people in Nebraska and Kansas. Each letter named a **target person** — a stockbroker in Boston — and asked the recipient to forward it to someone they knew *on a first-name basis* who might be "closer" to the target.

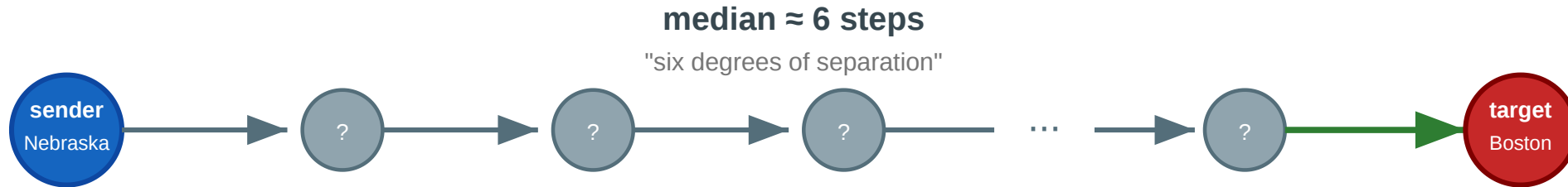
The letters that arrived took a **median of about 6 steps**.

The phrase "six degrees of separation" entered popular culture — but the experiment raised more questions than it answered.

# The Chain

## Milgram's small-world experiment (1967)

Can a letter reach a stranger in Boston through a chain of personal acquaintances?



### Questions this experiment raises

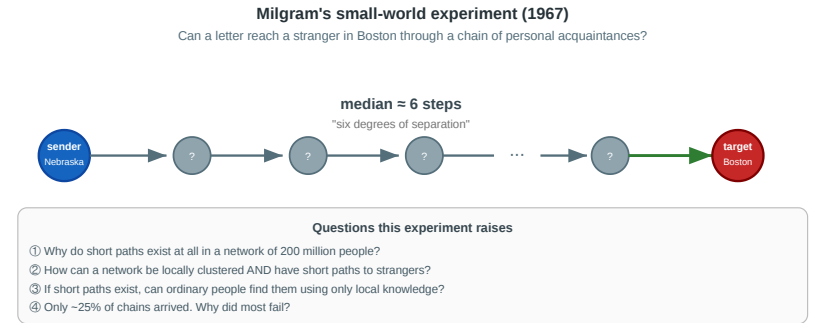
- ① Why do short paths exist at all in a network of 200 million people?
- ② How can a network be locally clustered AND have short paths to strangers?
- ③ If short paths exist, can ordinary people find them using only local knowledge?
- ④ Only ~25% of chains arrived. Why did most fail?

Each intermediary sees only their own contacts and must guess who is "closer" to the target. About 25% of chains completed; the rest died when someone refused to forward. The experiment demonstrates both that short paths *exist* and that finding them is *hard*.

# Four Questions from One Experiment

1. **Existence.** Why do short paths exist at all in a network of 200 million people?
2. **Coexistence.** How can a network be *locally clustered* (your friends know each other) and *globally short* (six steps to a stranger in another state)?
3. **Navigability.** If short paths exist, can ordinary people find them using only local knowledge — no map, no search engine?
4. **Failure.** Only ~25% of chains arrived. Does this mean short paths are rare, or that *finding* them is the bottleneck?

Every concept in today's lecture answers one of these.



# A New Kind of Gap — Existence vs. Findability

L03–L04 had a **computational** gap (NP-hard ideals). L05 had a **causal** gap. L06 had a **structural** gap (completeness required).

L07 has a **navigational** gap: short paths may *exist* without anyone being able to *find* them using local information.

| Lecture | Gap type      | Ideal                    | Reality                | Strategy                    |
|---------|---------------|--------------------------|------------------------|-----------------------------|
| L03–L04 | Computational | Exact optimization       | NP-hard                | Polynomial heuristics       |
| L05     | Causal        | Mechanism identification | Snapshots insufficient | Longitudinal data           |
| L06     | Structural    | Complete signed graph    | Sparse, noisy data     | Approximate balance         |
| L07     | Navigational  | Short navigable paths    | Local information only | Geometric link distribution |

# Takeaway

The small-world phenomenon is really two phenomena: the *existence* of short paths (explained by a few random long-range links) and the *findability* of those paths (explained only by a specific geometric distribution of links). Milgram's experiment revealed both — and the gap between them.

# The Small-World Phenomenon

**Guiding Question:** Why are distances in large networks often surprisingly short?



# Logarithmic Distances

**Small-world property.** A network exhibits the small-world property if the average shortest-path distance  $\bar{d}$  grows at most logarithmically with the number of nodes:

$$\bar{d} \propto \log |V|$$

In a random graph  $G(n, p)$  with average degree  $k$ , the typical distance is approximately  $\bar{d} \approx \frac{\log n}{\log k}$ .

# Logarithmic Distances

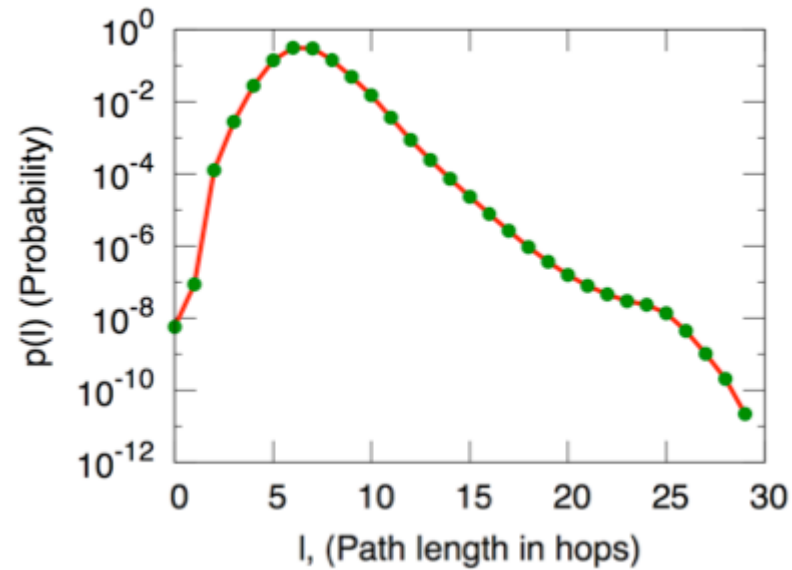
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For a social network of  $n = 300$  million people with average degree  $k \approx 100$ :  $\bar{d} \approx \frac{\log(3 \times 10^8)}{\log 100} \approx \frac{19.5}{4.6} \approx 4.2$ . Milgram's 6 is in the right ballpark.

# Empirical Examples



Source: Easley & Kleinberg 2010, p. Ch. 20

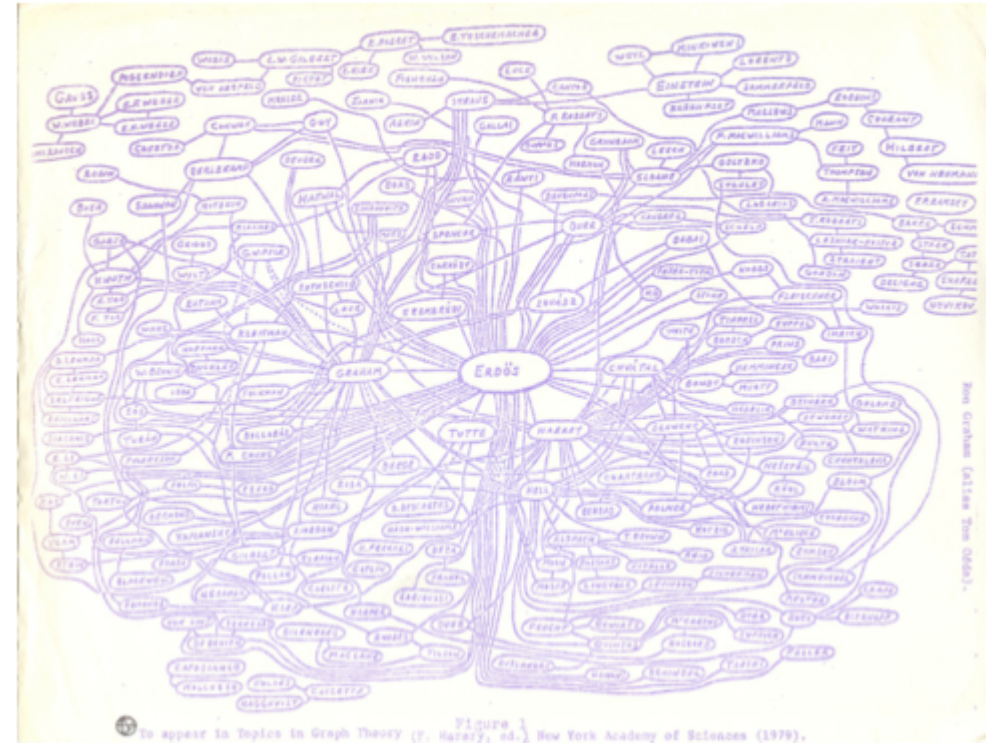


Figure 2.12: Ron Graham's hand-drawn picture of a part of the mathematics collaboration graph, centered on Paul Erdős [189]. (Image from <http://www.oakland.edu/enp/cgraph.jpg>)

Source: Easley & Kleinberg 2010

**Left:** Microsoft Instant Messenger network — 180 million users, median distance  $\sim 7$ . **Right:** Mathematical collaboration network — Erdős numbers rarely exceed 5. Both confirm logarithmic scaling across orders of magnitude.

# Takeaway

Logarithmic distances are a robust empirical fact across social, communication, and collaboration networks. In any network with moderate average degree, even billions of nodes remain only a handful of hops apart.

# Quick Check

The Facebook social graph has approximately 3 billion users and an average degree of about 300.

1. Estimate the average shortest-path distance using  $\bar{d} \approx \log n / \log k$ .
2. Facebook reported an empirical average distance of 3.57 in 2016. Is this consistent with your estimate?
3. Why might the empirical value be *lower* than the logarithmic estimate?

# Quick Check — Solution

$$\bar{d} \approx \frac{\log(3 \times 10^9)}{\log 300} \approx \frac{21.8}{5.7} \approx 3.8$$

Yes — 3.57 is remarkably close to the logarithmic estimate.

The  $\log n / \log k$  formula assumes homogeneous degree (random graph). Facebook has heavy-tailed degree — celebrities and organizations with millions of friends act as hubs, creating shortcuts that reduce  $\bar{d}$  below the random-graph prediction.

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# The Watts–Strogatz Model

**Guiding Question:** How can a network be both highly clustered *and* have short paths?

# The Paradox

Real social networks have two seemingly contradictory properties:

| Property             | Regular lattice      | Random graph                     | Real social network         |
|----------------------|----------------------|----------------------------------|-----------------------------|
| Clustering $C$       | High ( $\sim 0.5$ )  | Low ( $\sim k/n$ )               | High ( $\sim 0.1$ – $0.5$ ) |
| Avg. path length $L$ | Long ( $\sim n/2k$ ) | Short ( $\sim \log n / \log k$ ) | Short ( $\sim \log n$ )     |

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A regular lattice is clustered but large. A random graph is small but unclustered. **Real networks are both** — high  $C$  and low  $L$  simultaneously.

# The Model

Watts–Strogatz model (1998).

1. Start with a **ring lattice**:  $n$  nodes arranged in a circle, each connected to its  $k$  nearest neighbors.
2. For each edge, with probability  $p$ , **rewire** one endpoint to a uniformly random node (avoiding self-loops and duplicates).

The parameter  $p \in [0, 1]$  controls the interpolation:

- $p = 0$ : regular lattice (high  $C$ , high  $L$ )
- $p = 1$ : random graph (low  $C$ , low  $L$ )
- $0 < p \ll 1$ : **small-world regime** (high  $C$ , low  $L$ )

# The Characteristic Curves

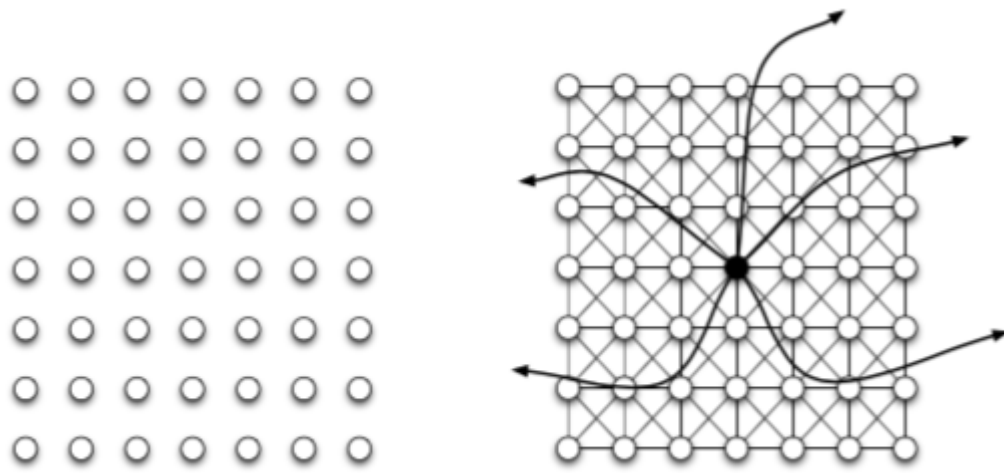


(a) *Pure exponential growth produces a small world*

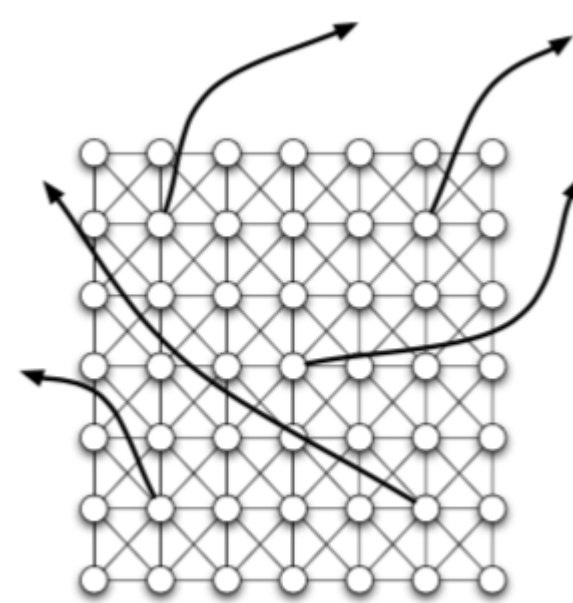


(b) *Triadic closure reduces the growth rate*

Source: Easley & Kleinberg 2010, p. Ch. 20



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$C(p)/C(0)$  stays near 1 while  $L(p)/L(0)$  drops sharply. The small-world regime (shaded) spans roughly  $0.001 \leq p \leq 0.1$  — a few percent of edges rewired is enough to make the world small.

# Takeaway

The Watts-Strogatz model resolves the clustering/path-length paradox: a tiny fraction of random long-range shortcuts collapses distances without destroying local structure. The small-world regime exists because  $L(p)$  falls much faster than  $C(p)$ .



# Quick Check

A ring lattice has  $n = 1000$  nodes and  $k = 10$  neighbors per node.

1. Estimate the average path length  $L$  at  $p = 0$ .
2. After rewiring 1% of edges ( $p = 0.01$ ), how many long-range shortcuts are created?
3. Qualitatively, what happens to  $L$  and  $C$ ?

# Quick Check — Solution

$L(0) \approx n/(2k) = 1000/20 = 50$  steps.

2.  $nk/2 = 5000$  edges total; 1% = .

3.  $L$  collapses — 50 random shortcuts in a 1000-node lattice bring  $L$  close to  $\log n / \log k \approx 3$ .  $C$  barely changes — only 50 of 5000 edges were rewired, so the vast majority of local triangles survive.

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# Kleinberg's Decentralized Search

**Guiding Question:** Short paths exist — but can local actors *find* them without a map?

# The Problem

In Milgram's experiment, each person forwarded the letter based on **local information only**:

- they knew their own contacts
- they knew the target's name, city, and profession
- they did *not* know the global network structure

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**Watts-Strogatz does not guarantee this works.** Short paths exist, but the greedy algorithm may not find them — it depends on *where* the long-range links point.

# Kleinberg's Grid Model (2000)

**Setup.** Place  $n$  nodes on a  $d$ -dimensional grid. Each node has:

- **local contacts:** all grid neighbors within lattice distance  $p$
- **one long-range link:** drawn with probability proportional to  $d(v, u)^{-r}$ , where  $d$  is grid distance and  $r \geq 0$  is the distance exponent.

**Greedy routing.** At each step, forward to the neighbor (local or long-range) closest to the target in grid distance.

The parameter  $r$  controls the *geometry* of long-range links:

- $r = 0$ : uniform random (long-range links go anywhere)
- $r = d$ : inverse-power law matched to dimension (links span all scales equally)
- $r \gg d$ : long-range links are effectively local

# The Three Regimes

## Kleinberg's grid model — the geometry of navigability

Each node has local grid neighbors + one long-range link drawn with  $P(\text{link to } u) \propto d(v,u)^{-r}$

$r = 0$  (uniform random)



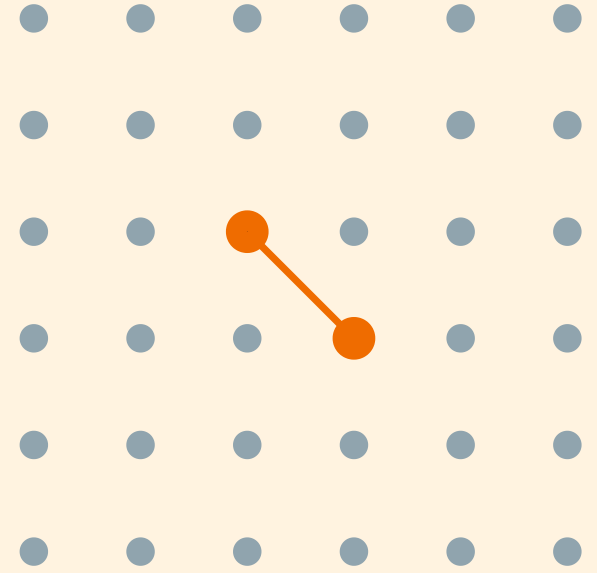
short paths exist,  
but search is  $\Omega(n^{2/3})$

$r = 2 = d$  (inverse-square)



short paths exist  
AND search is  $O(\log^2 n)$

$r \gg d$  (too clustered)



short paths exist,  
but "shortcuts" are too local

# Kleinberg's Theorem

Theorem (Kleinberg, 2000).

In the  $d$ -dimensional grid model with long-range exponent  $r$ :

- If  $r = d$ : greedy routing finds the target in  $O(\log^2 n)$  steps.
- If  $r \neq d$ : **no** decentralized algorithm (using only local information) can find the target in fewer than  $\Omega(n^c)$  steps for some constant  $c > 0$ .

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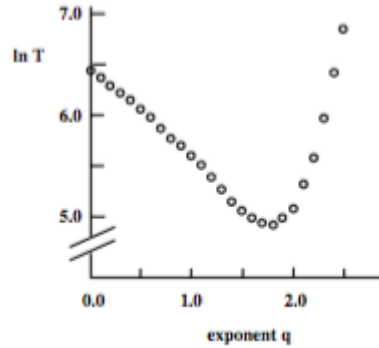
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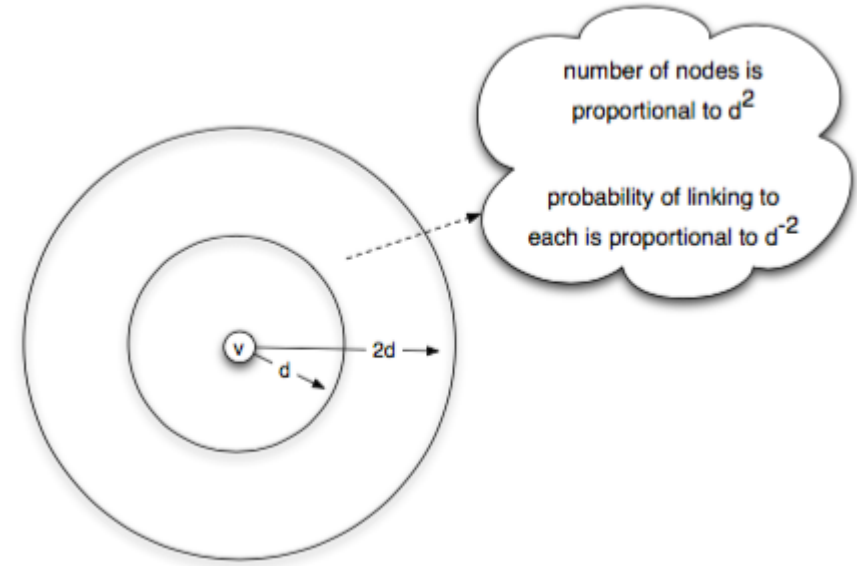
- If  $r = d$ : greedy routing finds the target in  $O(\log^2 n)$  steps.
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The exponent  $r = d$  is **unique** — it is the only value at which decentralized search is polylogarithmic. Any other value, and search degrades to polynomial.

# Simulation Evidence



Source: Easley & Kleinberg 2010, p. Ch. 20



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**Left:** delivery time vs. exponent  $r$  — a sharp minimum at  $r = d = 2$ . **Right:** the proof intuition — at  $r = d$ , each "ring" of nodes at distance  $2^i$  from the target receives roughly equal link probability, creating a hierarchy of scales the algorithm descends one level at a time.

# Takeaway

Short paths exist in almost any small-world network. But navigability — the ability of local actors to *find* those paths — requires a specific geometric distribution of long-range links ( $r = d$ ). Watts-Strogatz gives existence; Kleinberg gives findability.

# Quick Check

A social network is well-modeled by a 2D grid with random long-range links.

1. What exponent  $r$  makes greedy search efficient?
2. If the network's long-range links are uniformly random ( $r = 0$ ), what happens to (a) the existence of short paths and (b) greedy routing performance?
3. Why is the Watts-Strogatz model insufficient to explain Milgram's result?



# Quick Check — Solution

$r = 2 = d$  for a 2D grid.

- (a) Short paths — uniform random shortcuts reduce average path length.
- (b) Greedy routing — it takes  $\Omega(n^{2/3})$  steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ( $r = 0$ ). This gives short paths but not short paths. Milgram's participants actually found short paths using local information — that requires  $r = d$ , a much stronger structural condition.

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# Beyond the Grid — Where Else Does This Show Up?

Kleinberg's grid is a toy model. But the *principle* — mix **short-range** links (for local refinement) with **long-range** links at every scale (for jumping) — generalises far beyond social networks.

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The same small-world recipe powers **modern vector search**: the index graphs used inside Pinecone, Weaviate, Milvus, FAISS, and every retrieval-augmented LLM pipeline are Kleinberg's grid in disguise.

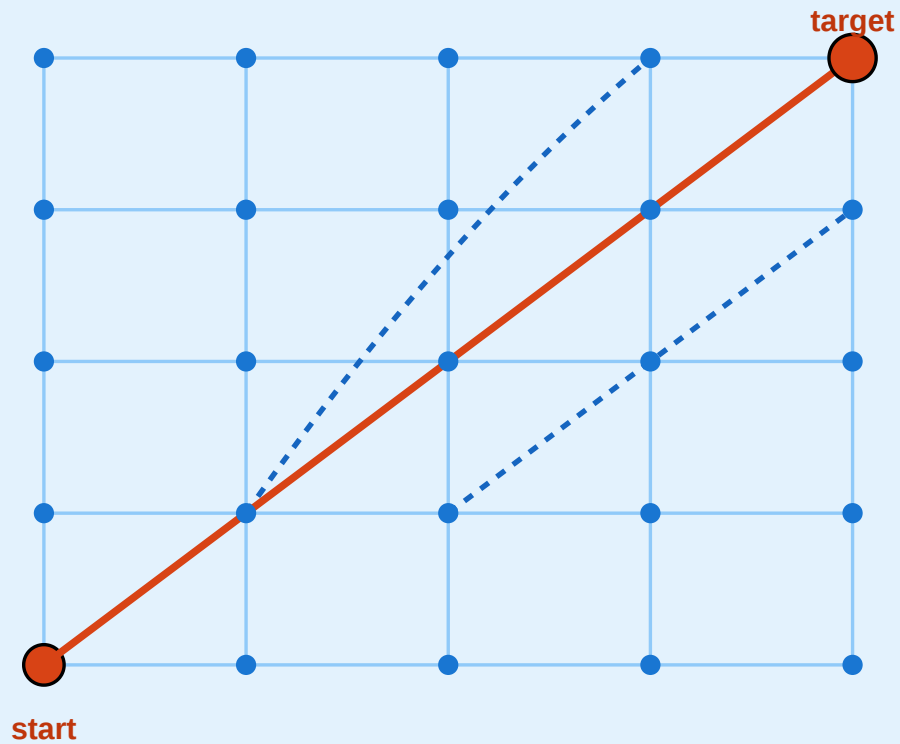
# From Kleinberg's Grid to HNSW

# From Kleinberg's Grid to Modern Vector Search (HNSW)

Same principle: long-range + short-range links enable greedy routing

## Kleinberg 2000 — Geometric Grid

known 2-D coordinates ·  $r = d = 2 \cdot O(\log^2 n)$

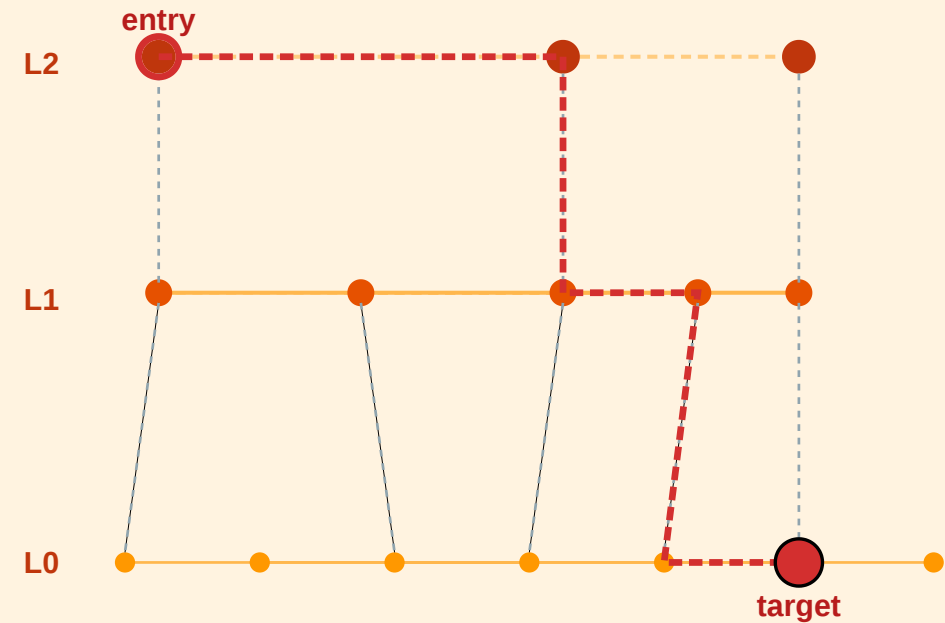


known metric · distance =  $|u - v|$  in the grid

generalise  
metric  
→ high-dim  
vectors

## HNSW — Hierarchical Navigable Small World

learned vector space · layered graph ·  $O(\log n)$



top layer = long-range jumps

bottom layer = local refinement

# HNSW in Three Ideas

- 1. Layers as scales.** Each node is assigned a maximum level  $l \sim \text{Geom}(1 / \ln M)$ . Layer  $k$  contains only nodes with level  $\geq k$  — so higher layers are exponentially sparser.
- 2. Greedy per layer.** Search starts at the top-layer entry point, greedily moves to the neighbour closest to the query, then descends one layer and repeats — refining locally.
- 3. Long-range by construction.** Top-layer edges span the whole space (like Kleinberg's  $r = d$  links); bottom-layer edges are local. The geometric distribution is *baked into the index*, not drawn from a random process.

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**Complexity.**  $O(\log n)$  query time in practice; billions of vectors routinely indexed in production.

# Why This Matters for RAG and LLM Retrieval

Every retrieval-augmented generation (RAG) pipeline has the same structure:

1. **Embed** each document chunk into  $\mathbb{R}^d$  (typically  $d = 384-4096$ ).
2. **Index** the vectors in an HNSW (or IVF, or DiskANN) graph.
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We will come back to **how** these embeddings are constructed in **Lecture 09** (node and document representations, spectral methods, DeepWalk, node2vec, GNNs). The *geometry* of the embedding space determines whether navigation works — which closes the loop to Kleinberg.



# Empirical Anchor: The Global Email Experiment

**Guiding Question:** Does Milgram's result replicate at scale — and what does chain failure tell us?

# The Study

Dodds, Muhamad & Watts (2003), *An Experimental Study of Search in Global Social Networks*, [Science 301, 827–829](#).

They replicated Milgram's experiment using **email** — a global medium with no geographic constraint.

| Parameter                       | Value                      |
|---------------------------------|----------------------------|
| Participants recruited          | ~60 000 from 166 countries |
| Target persons                  | 18 in 13 countries         |
| Chains started                  | 24 163                     |
| Chains completed                | 384 (~1.6%)                |
| Median chain length (completed) | 4–7 steps                  |

# What They Found

- 1. Completed chains are short.** Median 4–7 steps, consistent with Milgram and with  $\log n / \log k$  scaling.
- 2. Most chains die early.** The completion rate of ~1.6% is not because paths don't exist — it's because intermediaries *choose not to forward*. The bottleneck is **motivation**, not topology.

# What They Found

- 1. Completed chains are short.** Median 4–7 steps, consistent with Milgram and with  $\log n / \log k$  scaling.
- 2. Most chains die early.** The completion rate of ~1.6% is not because paths don't exist — it's because intermediaries *choose not to forward*. The bottleneck is **motivation**, not topology.
- 3. Successful chains use geography and profession as search dimensions.** Forwarders who brought the letter closer in *geographic* or *professional* distance were more effective — consistent with Kleinberg's model where search exploits multiple distance dimensions simultaneously.

# What the Study Could Not See

- 1. Chain death  $\neq$  path absence.** A chain that dies after 3 steps does not mean no short path existed — it means someone chose not to forward. The study measures *willingness to search*, not *path existence*. The true completion rate under full cooperation is unknown.
- 2. Selection bias in completions.** The 384 completed chains are a biased sample — they may systematically over-represent easy targets (well-known, geographically close, professionally distinctive). The median chain length of 4–7 steps may underestimate the true difficulty of reaching a random target.
- 3. Email  $\neq$  acquaintance.** Milgram's experiment required *personal acquaintance*; the email study allowed forwarding to anyone whose email address you knew. The social network traversed by email may be larger and weaker-tied than the face-to-face network Milgram assumed.
- 4. No ground-truth network.** Without observing the full network, the study cannot measure whether the successful chains actually followed near-optimal paths or just happened to complete.

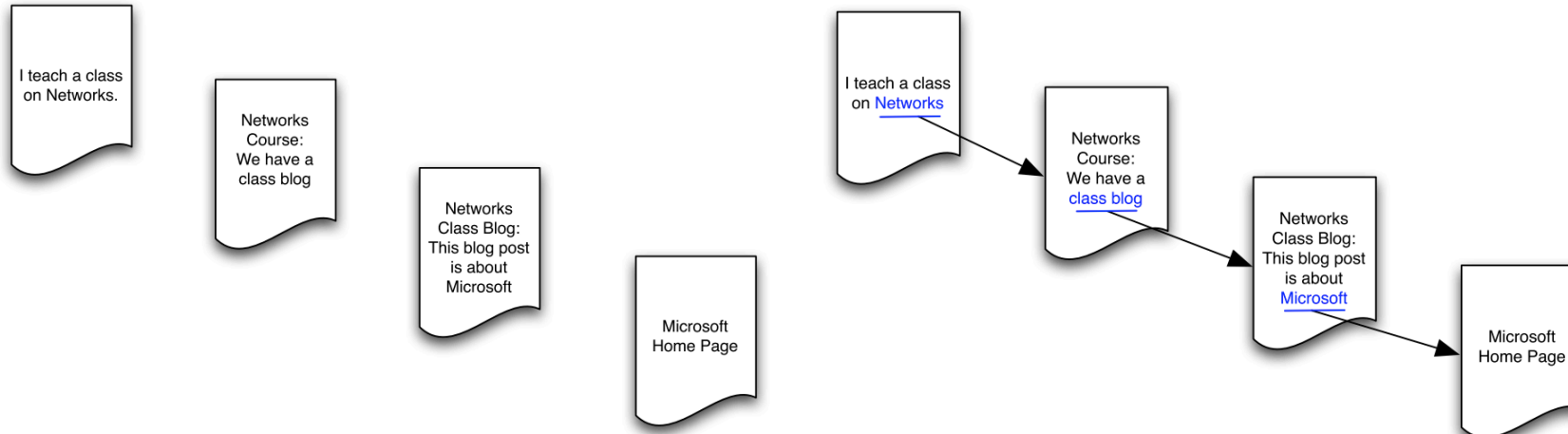
# Takeaway

Dodds et al. (2003) confirmed Milgram at scale: short paths exist and people can sometimes find them. But the 1.6% completion rate reveals that *motivation*, not topology, is the primary bottleneck. The navigational gap is real — short paths exist but are rarely found in practice.

# The Web as a Directed Small World

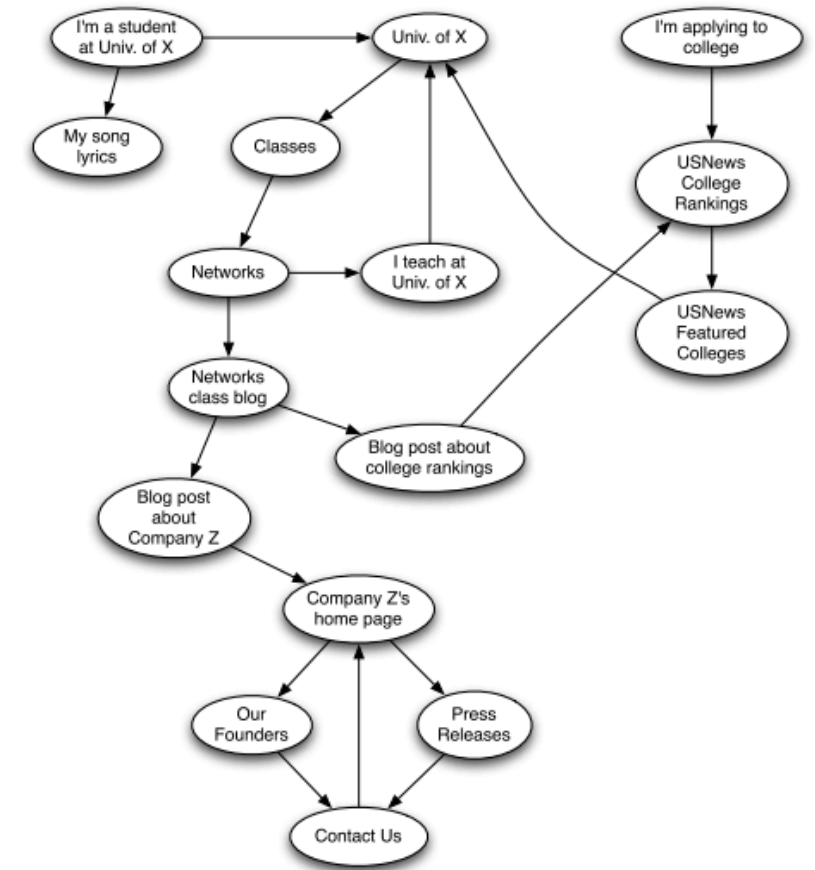
**Guiding Question:** What happens to navigability when links are directed?

# From Social to Information Networks



Source: Easley & Kleinberg 2010

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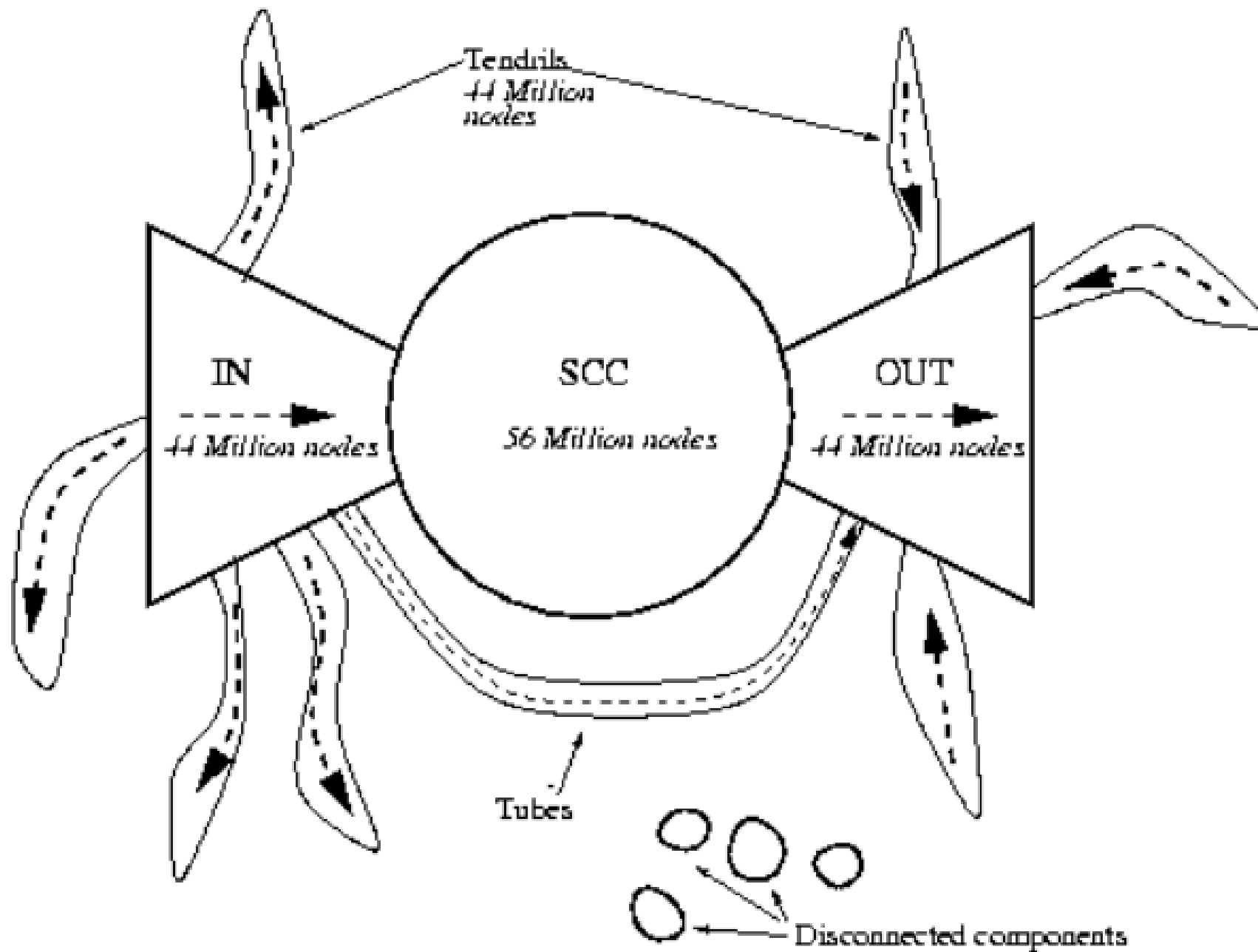
Source: Easley & Kleinberg 2010

Hyperlinks are **directed** — page A can link to page B without B linking back. This means reachability is asymmetric: you can follow links *from A to B* but not necessarily in reverse.



# The Bow-Tie Structure





Source: Easley & Kleinberg 2010, p. Ch. 13

# Navigability in Directed Networks

The small-world property and navigability both change in directed networks:

- **Short paths exist** within the SCC — the strongly connected core is a small world (~16 clicks between random page pairs within SCC).
- **Short paths do not exist** from OUT back to SCC, or between tendrils.
- **Greedy search is harder** because you can only follow outgoing links — you cannot "pull" information from a page that doesn't link to you.

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**Search engines** (Google, 1998) solve the navigability problem by building a *global index* — they observe the entire graph and compute optimal paths. This is the opposite of Kleinberg's decentralized search: it works precisely because it is *centralized*.

# Takeaway

The Web's directed links create asymmetric reachability. Only the Strongly Connected Component (~28%) is a "small world." Search engines solve the navigability problem through centralized indexing — the opposite of Milgram-style decentralized routing.

# Quick Check

A directed network has 1000 nodes. The SCC contains 300 nodes. The IN component has 200 nodes, and the OUT component has 250 nodes. The rest are tendrils.

1. From a random node in IN, can you reach a random node in OUT? Explain your path.
2. From a random node in OUT, can you reach a node in IN?
3. What fraction of all possible source-target pairs have a directed path between them?

# Quick Check — Solution

IN  $\rightarrow$  SCC  $\rightarrow$  OUT. By definition, IN nodes can reach SCC, and SCC nodes can reach OUT. The path goes through the SCC as a relay.

OUT nodes cannot reach back to the SCC (that is what defines them as OUT, not SCC). So OUT  $\rightarrow$  IN has no directed path.

**3. Reachable pairs include:** SCC $\rightarrow$ SCC ( $300 \times 299$ ), IN $\rightarrow$ SCC ( $200 \times 300$ ), SCC $\rightarrow$ OUT ( $300 \times 250$ ), IN $\rightarrow$ OUT via SCC ( $200 \times 250$ ). That totals  $\sim 240,000$  pairs out of  $1000 \times 999 \approx 10^6$  — roughly 24%. Most pairs are mutually reachable.

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# Wrap-Up — The Five Gaps

| Lecture     | Gap type      | Ideal                    | Reality                         | Strategy                                        |
|-------------|---------------|--------------------------|---------------------------------|-------------------------------------------------|
| L03–<br>L04 | Computational | Exact optimization       | NP-hard                         | Polynomial heuristics                           |
| L05         | Causal        | Mechanism identification | Snapshots insufficient          | Longitudinal data                               |
| L06         | Structural    | Complete signed graph    | Sparse, noisy data              | Approximate balance                             |
| L07         | Navigational  | Short navigable paths    | Local info only; directed links | Geometric link distribution; centralized search |

The small-world phenomenon says short paths *exist*. Watts-Strogatz explains *why*. Kleinberg explains *when* they are *findable*. Milgram and Dodds et al. tested *whether* real people find them — and the answer is: sometimes, with effort, and only when the network's geometry cooperates.

**Looking Ahead: Lecture 08**



How do things *spread* on networks — diseases, information, behaviors? The structure we have built (ties, communities, balance, small-world shortcuts) now becomes the substrate for dynamic processes: contagion, diffusion, and cascading failure.

## Reading

Chapter 20 of Easley & Kleinberg (2010) covers the small-world phenomenon and Watts-Strogatz. Kleinberg (2000), *The Small-World Phenomenon: An Algorithmic Perspective*, is the source for the decentralized search theorem. Dodds, Muhamad & Watts (2003) is the email experiment. Broder et al. (2000) is the bow-tie structure of the Web.

## Image Credits & References

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning about a Highly Connected World. *Cambridge University Press*. <https://www.cs.cornell.edu/home/kleinber/networks-book/>